

From: Yardeni QuickTakes <yardeni-research@ghost.io>
Subject: Global Bond Rout Made In Japan?
Date: 1 hour ago



YARDENI QUICKTAKES

Global Bond Rout Made In Japan?

By Ed Yardeni & Elias Griepentrog • 17 Sept 2026

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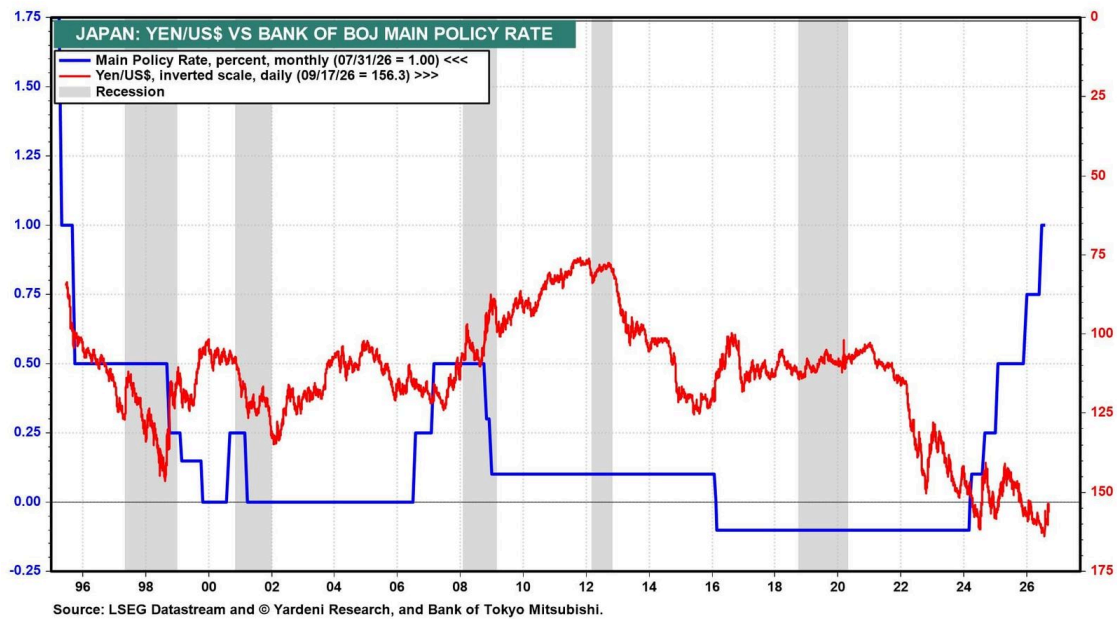
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I. Yen-Carry Trade Unwinding?

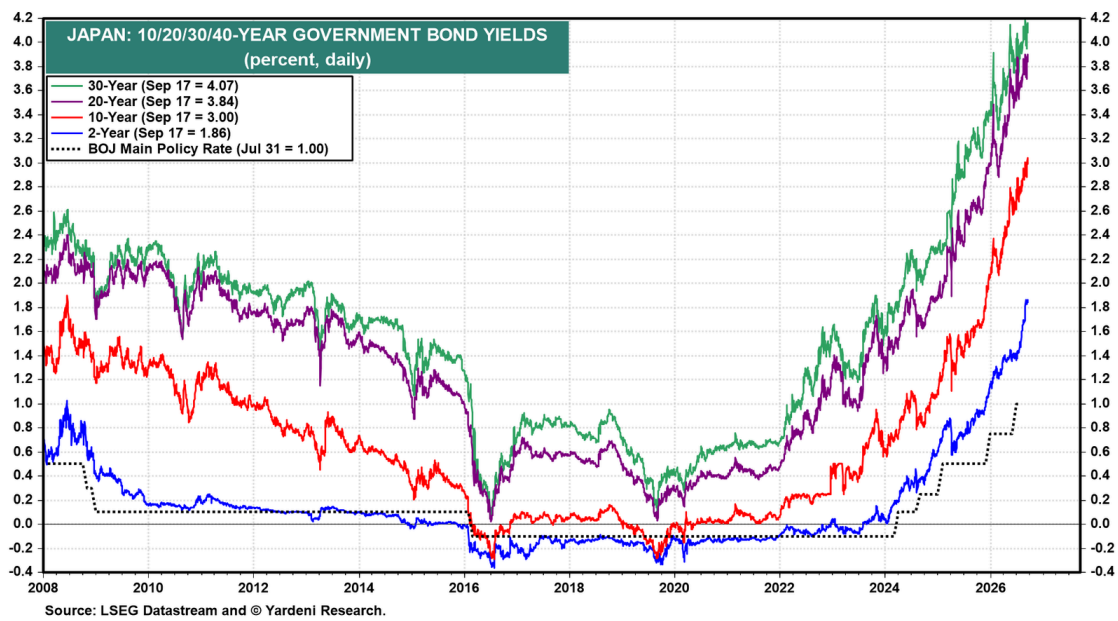
The “yen-carry trade” has been a key feature of global financial markets since roughly 2012. It rested on two pillars: ultra-low Japanese interest rates and either a weak or relatively stable yen. Hedge funds could borrow funds cheaply in yen, convert the proceeds to other currencies, and buy government bonds in those currencies. The beauty of this trade is that it increased downward pressure on the yen as long as the Bank of Japan (BOJ) kept its official policy rate near zero (chart).

Today, both pillars are cracking. Since early 2024, after years of near-zero and even negative rates, the BOJ has raised its official policy rate to 1.0%, the highest since 1995, with another 25bps hike expected tomorrow morning. Meanwhile, the yen has become more volatile and is expected to strengthen in response to tighter monetary policy. After weakening to around ¥163 per dollar, near a four-decade low, it has rallied since late July following joint Japan-US intervention in the forex market.

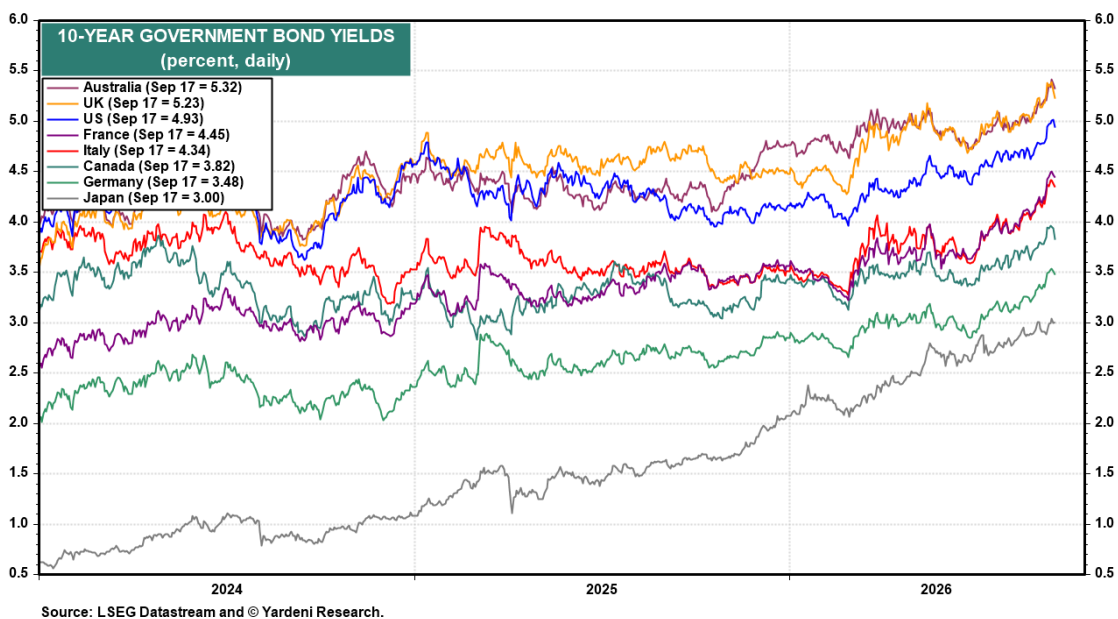
Higher Japanese interest rates and likely further yen appreciation have been forcing traders to unwind their yen-carry trades. This might explain the rout in the global bond market since 2024.



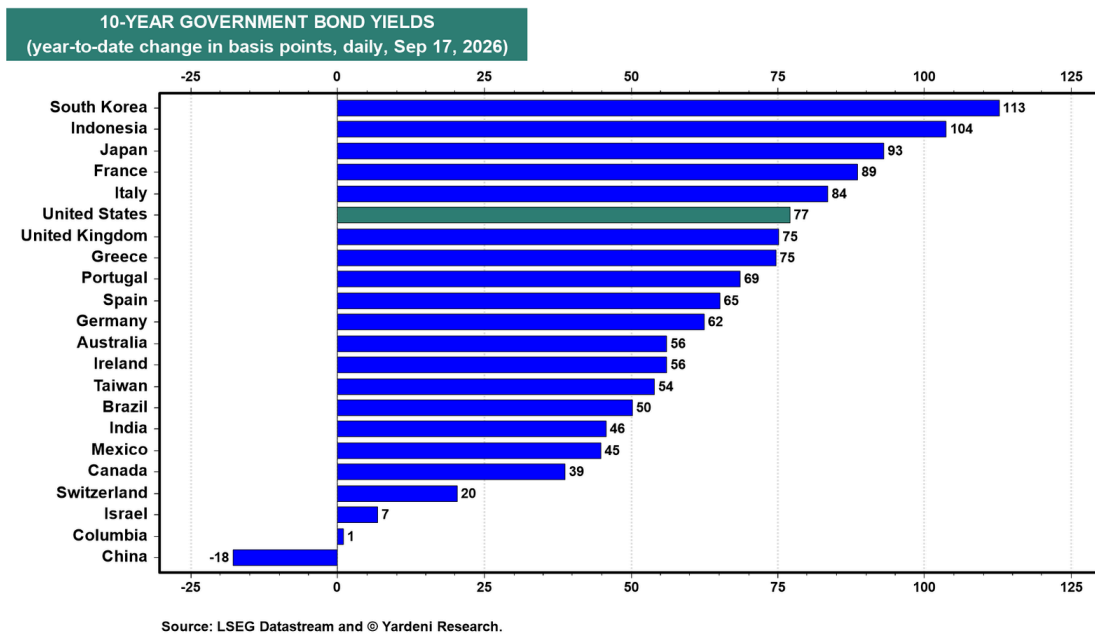
Japan's bond market confirms the BOJ has more tightening to do. JGB yields have risen sharply alongside the policy rate but remain well above it across the curve (chart). Japan's Bond Vigilantes are signaling that monetary policy remains too accommodative. Higher Japanese bond yields also encourage Japanese bond investors to return home and reduce their exposure to foreign bonds, especially if the yen continues to rally.



The surge in the 10-year JGB yield has occurred alongside a broad rise in global government bond yields (chart). In our view, the unwinding of yen-funded positions may be a key contributor to the synchronized rise in these bond yields.

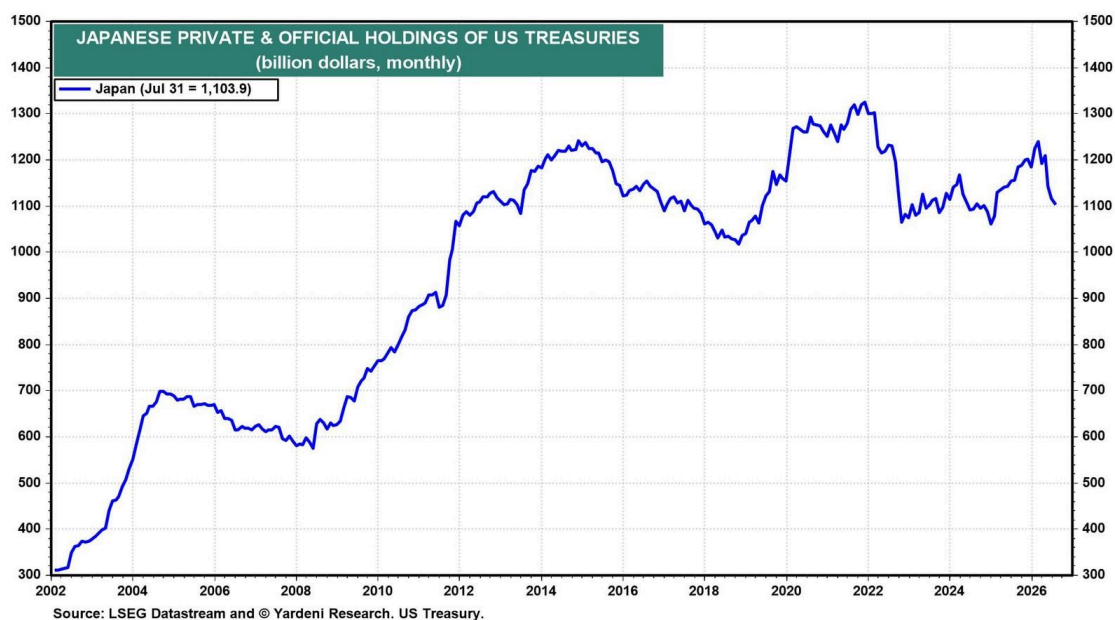


Japan is leading the global bond selloff. Its 10-year yield is up 93bps this year, one of the largest increases globally (chart). We will be interested to see how the BOJ's rate decision affects global yields tomorrow.

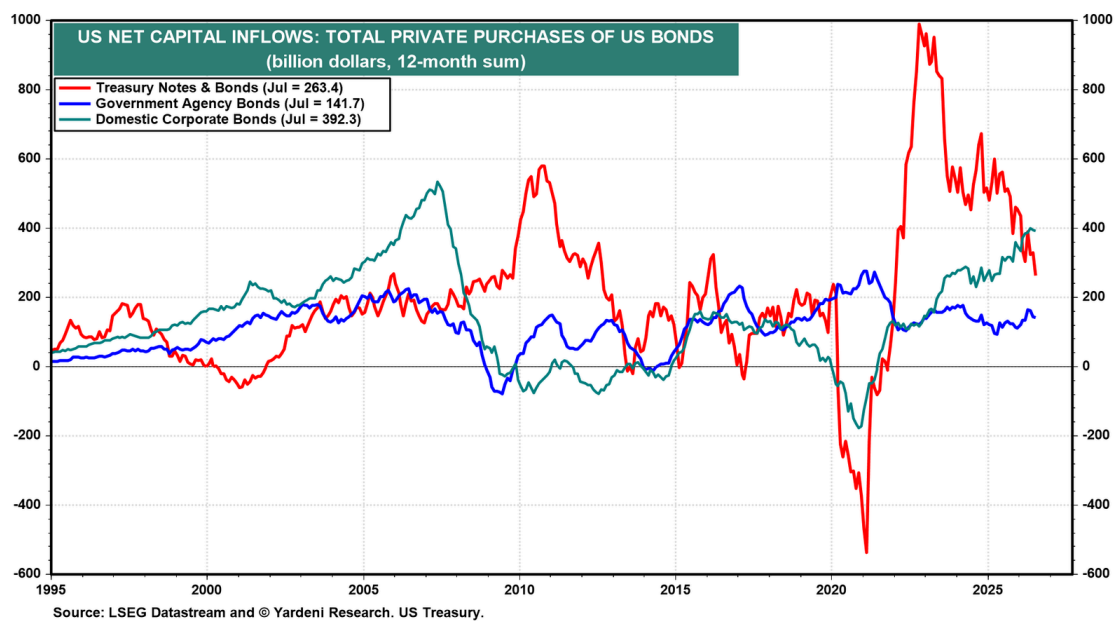


II. US Capital Flows

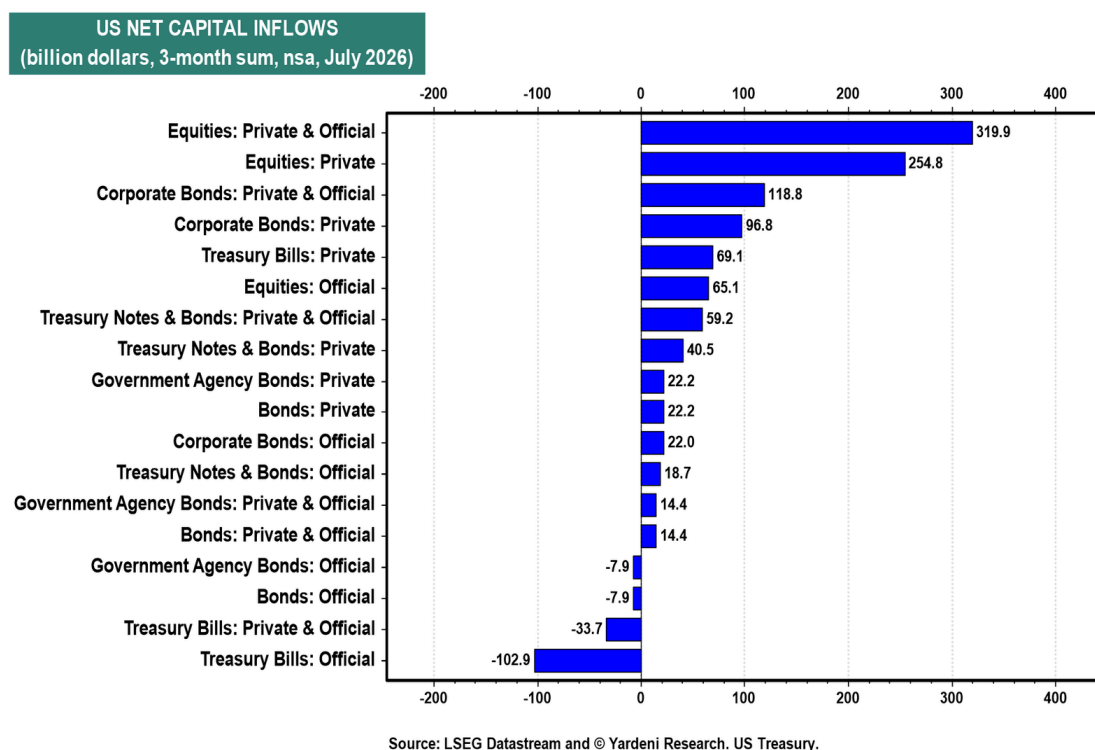
Holdings of US Treasuries by all Japanese accounts (both private and official) have declined recently and appear to be trending lower (chart). The Japanese may be unwinding their overseas positions in global bonds too, as domestic yields rise, making JGBs more attractive again.



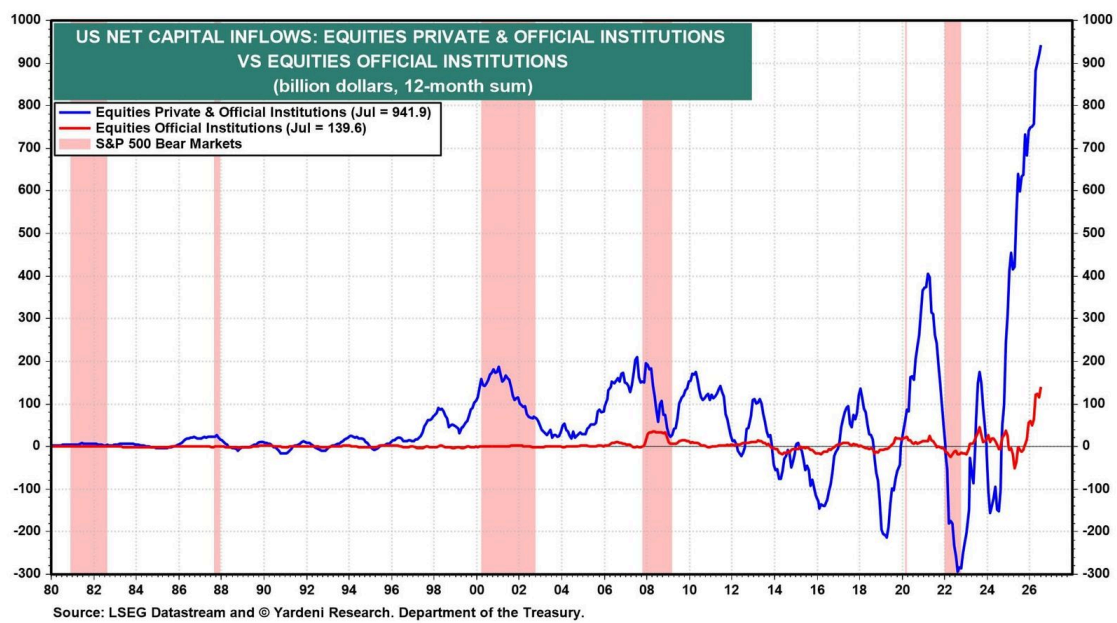
Total private foreign purchases of US Treasury notes and bonds fell to \$263.4 billion over the past 12 months, the lowest since 2022. Purchases of US corporate bonds totaled \$392.3 billion over the past 12 months, as foreign investors increasingly favor investment-grade debt tied to the AI buildout (chart).



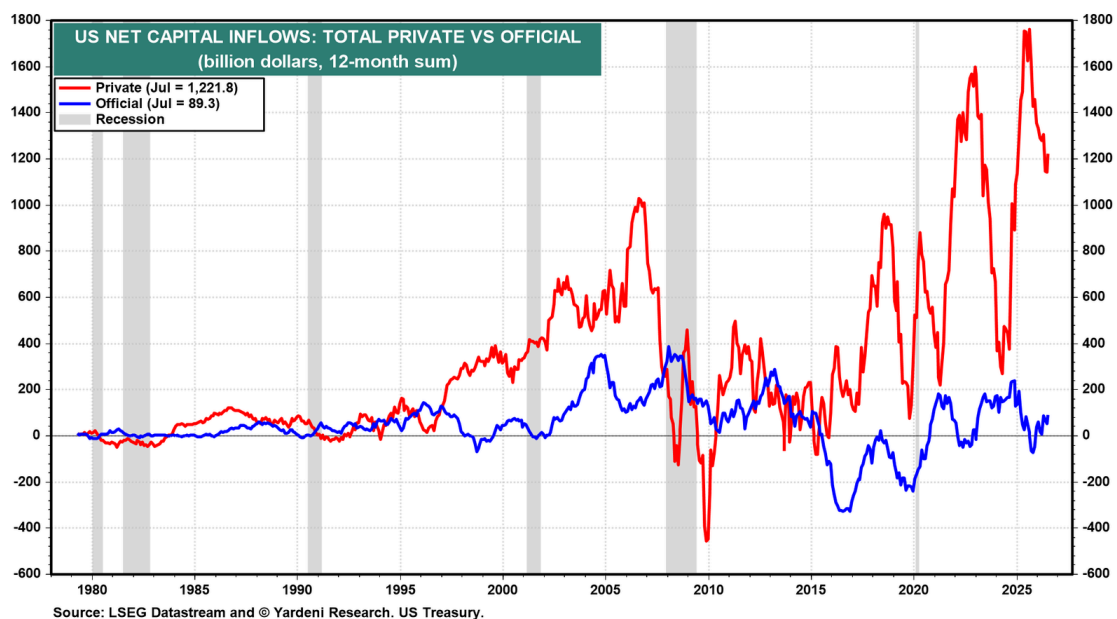
The shift in the composition of foreign demand for US assets is also clear over the past three months through July. Equities attracted the largest inflows, followed by corporate bonds, while purchases of Treasury notes and bonds were much smaller (chart).



Now get this: Over the past 12 months, foreigners purchased a record \$941.9 billion in US equities (chart)! This includes \$139.6 billion in US equity purchases by foreign official accounts over the past 12 months.



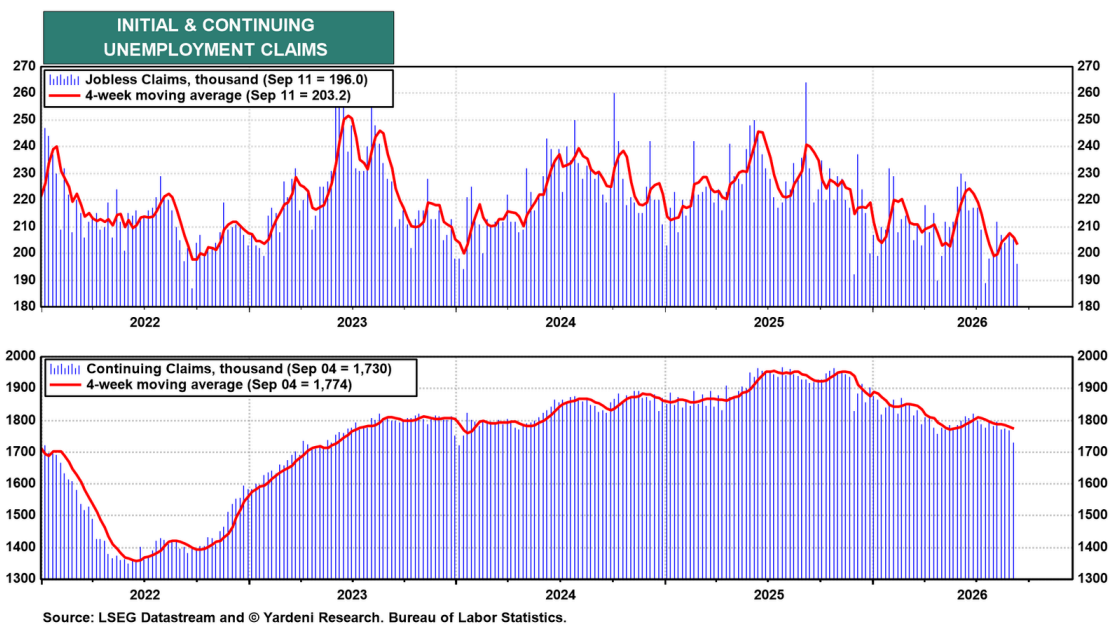
In aggregate, private net foreign capital inflows into the US remained elevated at around \$1.2 trillion over the past 12 months (chart). Net inflows from foreign official accounts totaled just \$31.7 billion.



III. US Economic Indicators

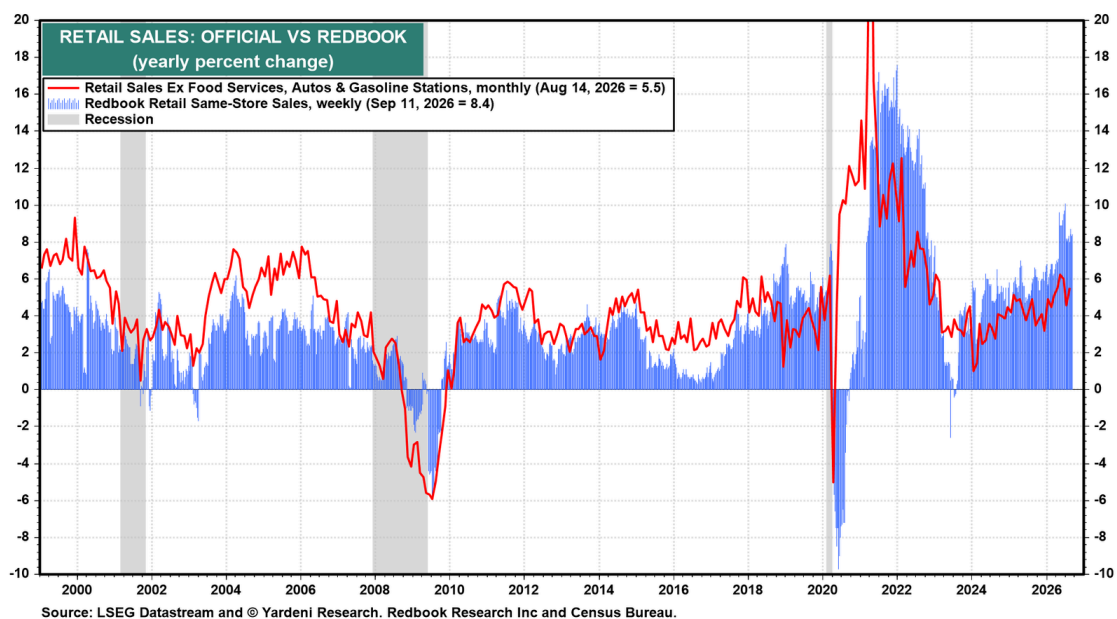
The strength of the US economy remains the key reason private foreign inflows into US equities and corporate bonds are so strong. Here is a look:

(1) Jobless claims. The US labor market remains in good shape. Initial jobless claims fell to 196,000 during the week of September 11 and have now come in below 200,000 five times this year, versus just once in 2025 (chart). Meanwhile, the four-week moving average of continuing claims fell to its lowest level since January 2024 and has declined for four consecutive weeks.



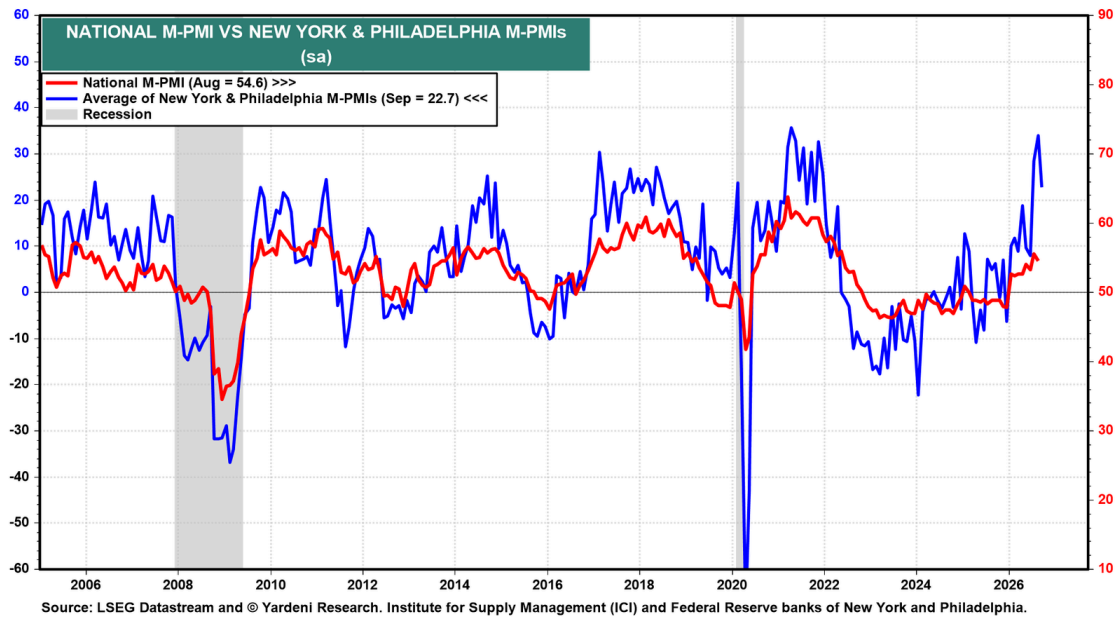
(2) Consumer spending. After the August retail sales report showed consumer spending remained robust, Redbook data suggest that strength has carried into September. Same-store sales rose 8.4% y/y during the week of September 11, well above the 5.8% average in 2025 (chart).

Bank of America's August [Consumer Checkpoint Survey](#) also points to robust spending. Card spending per household rose 0.9% m/m and 4.5% y/y, more than four times the 2025 average. Excluding gasoline, spending rose 3.7% y/y, more than 2.5 times the 2025 pace.

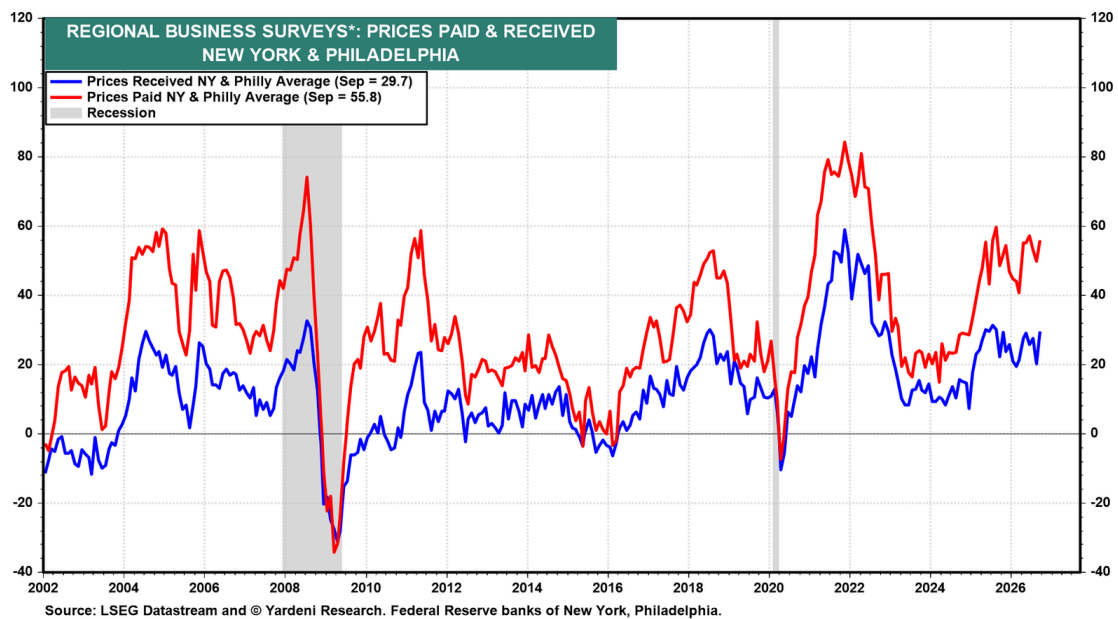


(3) Manufacturing. Economic activity in manufacturing also remains remarkably robust. The average of the New York and Philadelphia Fed manufacturing indexes remained elevated at 22.7 in September,

suggesting the national M-PMI likely remained comfortably in expansion territory (chart).



The regional prices-paid and prices-received indexes remained high, suggesting that inflation pressures remain troublesome (chart).



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Subject: 17-21% On-demand AI GPU pricing hike to shift value from LLM companies to cloud vendors
Date: 5 hours ago

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17-21% On-demand AI GPU pricing hike to shift value from LLM companies to cloud vendors

Nebius 10/1 全面調漲 Nvidia AI GPU, CPU, Memory 租用價格，我們認為短期市場價值從賣 tokens 的大語言/專家語言模型公司轉到 AI 數據中心算力基礎建設公司

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What's new?

根據路透社 ([Nebius hikes AI cloud prices again as demand for computing power soars](#))，Euronext, MarketBeat 9 月 17 日報導，Nebius Group 宣布自 2026 年 10 月 1 日起局部調升 On-demand 算力租金：其中 Nvidia H100 租用價格由每小時 US\$3.85 調升 16.9% 至 US\$4.50；次世代旗艦 B300 算力租金更大漲 21% 從每小時 US\$7.85 調升到 US\$9.50。除此以外，Nebius 也調高部分 AMD/Intel x86 CPU 租用價格達 25%，及存儲記憶體租用價格達 41%，Nebius 同步確認其簽約容量全速朝向年底 5 GW 目標邁進。同日，CoreWeave 官方宣布自 6 月底以來持續以更高價格簽約算力，其第三季度簽訂的 3 至 6 個月短期客戶合約，年化定價達到每 MW 約 4,000 萬美元（折合 1 GW 算力租用合約每年高達 400 億美元），合約總用電容量自 3.7 GW 增至 4.2 GW。為支撐龐大建置，CoreWeave 同步啟動 30 億美元可轉換優先票據發售（另附 5 億美元超額配售權），並設立高達 3,500 萬股（約 29.2 億美元）的市價增發（ATM）融資機制。



付費訂閱戶可分享我們對於受惠，受影響產業鏈分析，
淨負債及4年累積營收/在手訂單比較表，及產業建議。

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Positively affect what supply chain vendors

- 前沿 GPU 與高階晶片領導者 (NVIDIA、台積電)：Nebius 對 H100, B300, x86 CPU, 存儲記憶體算力租用漲價且客戶照單全收，證實目前AI 算力租用仍在賣方市場，AI算力晶片仍具備絕對賣方定價權，對於外溢雲 Neocloud 如Oracle, Nebius, CoreWeave, SpaceXAI 算力租用的毛利率，及營業利潤率結構會持續改善而高於市場預期（請參考7大算力數據中心Cloud 業務營業利潤率比較表），當然這明顯也會擴散到提供 ASIC/TPU 算力租用的亞馬遜及谷歌，及有利於AI GPU晶片龍頭 Nvidia, x86 CPU 巨頭AMD, TPU 龍頭谷歌，及先進晶圓代工台積電營收及獲利的增量改善。
- 資料中心高壓電氣與關鍵冷卻設備商 (Eaton、Schneider Electric、Vertiv)：CoreWeave 簽約容量推升至 4.2 GW，年化每 MW 租金衝上 4,000 萬美元，激勵機房電力建設廠商享受極高交期溢價。

	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26	4Q26
Cloud OP%								
Google Cloud	18%	21%	24%	30%	33%	36%	33%	33%
AWS/Amazon	40%	33%	35%	35%	38%	39%	36%	36%
Intelligent Cloud/Microsoft	41%	41%	43%	42%	40%	41%	42%	41%
Oracle Cloud	31%	32%	29%	29%	32%	32%	35%	32%
SpaceXAI Cloud		-207%			-302%	-49%	2%	22%
Coreweave CRWV	-3%	2%	4%	-6%	-7%	-2%	0%	5%
Nebius NBIS	-234%	-106%	-89%	-103%	-32%	-30%	-28%	-21%
Top 7	35%	31%	35%	35%	33%	35%	35%	34%
Overall OP%								
Google	34%	32%	31%	32%	36%	34%	34%	34%
Meta	41%	43%	40%	41%	41%	31%	32%	34%
Amazon	12%	11%	10%	12%	13%	14%	13%	13%
Microsoft	46%	45%	49%	47%	46%	45%	48%	47%
Oracle	31%	32%	29%	29%	32%	32%	35%	32%
SpaceXAI	1%	-24%			-41%	-2%	18%	28%
Coreweave CRWV	-3%	2%	4%	-6%	-7%	-2%	0%	5%
Nebius NBIS	-234%	-106%	-89%	-103%	-32%	-30%	-28%	-21%
Top 8	27%	27%	26%	26%	28%	27%	27%	27%

Source: The companies, Andrew Lu on global semis/techs, Bloomberg est.

Negatively affect what supply chain vendors

資金鏈脆弱的二線 AI 模型訓練新創：短期突發AI GPU算力租金價格大漲達 17-21%，x86 CPU 租用價格跳升達25%，存儲記憶體租用價格飆升達 41%，每GW算力年化達 400億美元，大幅拉高訓練中繼或追加推論負載的邊際成本，現金消耗速度失控，對於一線大語言模型公司 Anthropic 及 OpenAI 還有龐大付費訂閱及Tokens 銷售營收ARR可以支撐，但對於缺乏龐大訂閱用戶及Tokens 銷售的二線大語言及專家模型新創公司就相當辛苦，尤其是每百萬Tokens 價格受到中國大陸開源模型競爭而持續下滑。

Our outside the box non-consensus thinking:

我們之前常將 Oracle 及 CoreWeave 龐大負債視為「流動性吃緊的危險信號」（請參考淨負債對股東權益比較下方表格）。非共識深度洞察在於，這是外溢雲 Neocloud 巨頭在主動實施以龐大算力租用在手訂單需求（請參考下圖，Till 2Q26 各大AI算力租用數據中心在手訂單是未來4年預測營收加總的10倍左右）的強大支撐來反應AI 算力數據中心基礎建設，電力及晶片價格價格的飆漲。短期 3-6 個月合約能賣到每 MW 年化 4,000 萬美元，代表市場極度缺乏能夠「立即通電」的即戰力算力。AI算力數據中心趁合約定價處於歷史高檔，以可轉債鎖定低成本資金，是在為 2027 年可能出現的電力擴建壁壘建立不可跨越的護城河。

	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26
Net debt to equity ratio							
Google	-20%	-16%	-16%	-15%	-6%	-19%	
Meta/FB	-11%	1%	3%	2%	2%	8%	
Amazon	18%	18%	16%	11%	19%	22%	
Microsoft	8%	5%	5%	9%	11%	12%	
Oracle	455%	466%	383%	344%	315%	315%	197%
SpaceXAI total					16%	-48%	
Coreweave CRWV	346%	351%	435%	799%	691%	917%	
Nebius NBIS	-40%	-39%	-6%	28%	3%	19%	
Alibaba	-49%	-50%	-45%	-48%	-44%	-42%	
Tencent	2%	3%	-1%	-2%	-5%	1%	

Source: Bloomberg, Andrew on global semis/techs

	2Q25	3Q25	4Q25	1Q26	2Q26	3Q26
4yrs incremental sales to backlog						
Google	47%	37%	26%	10%	10%	12%
Amazon AWS	24%	25%	22%	14%	11%	
Microsoft	14%	14%	9%	9%	9%	
Oracle	27%	8%	7%	7%	7%	5%
CoreWeave	39%	24%	24%	10%	9%	8%
Total	23%	17%	13%	10%	9%	17%

Source: The companies, Andrew Lu on global semis/techs

Our advises to investors

Nebius 10/1 全面調漲 Nvidia AI GPU, CPU, Memory 租用價格, 我們認為短期市場價值從賣 tokens 的大語言/專家語言模型公司轉到 AI 數據中心算力基礎建設公司及其產業鏈, 尤其是具備自建電力與獨立變电站能力的重資產 AI 算力數據中心, AI GPU/TPU, CPU, memory 龍頭; 持續做多具備定價權的半導體晶圓代工龍頭 (台積電); 對無固定電力配額的純二手轉租商進行折價。但我們還是要密切關注超大規模雲巨頭 (微軟、亞馬遜, 谷歌) 的 AI 算力需求變化, 是否會釋出閒置算力打擊現貨市價。

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Is CPO technology killing M8/M9 PCB CCL demand?

CPO 技術未來是否將嚴重影響 M8/M9 等級超低耗損銅箔基板 CCL 需求？

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What's new?

CPO (Co-Packaged Optics, 共封裝光學) 讓主機板 **PCB** 的銅箔基板 (CCL) 等級得以從極低損耗的 **M8 / M9 (Extreme Low Loss)** 大幅降階至 **M4 (Mid/Low Loss)**，其根本原因在於高速電信號在 **PCB** 銅箔上的傳輸距離 (**Trace Length**) 被壓縮至接近於零，傳輸損耗不再由主機板承擔。但從投資或產業鏈的視角來看，未來 1-2 年 (至 2H28 年) 無需擔憂 **CPO** 衝擊 **M8/M9** 需求，因為 **224G** 插拔模組與大型 **AI** 叢集的出貨量正在迎來陡峭爬坡期，但真正需要警戒的時間窗口在 **2028 年底至 2029 年**。

Why?

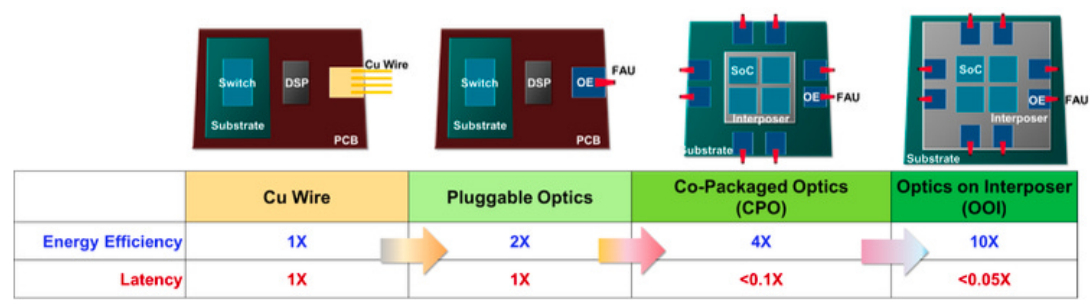
1. 傳輸通道長度與介質損耗的物理本質

高頻高速電信號 (如 112G PAM4、224G PAM4) 在 PCB 上的損耗，主要取決於兩大因素：

- **介質損耗 (Dielectric Loss)**：正比於傳輸頻率、材料介電耗損因子，以及走線長度。
- **導體表面粗糙度與集膚效應 (Skin Effect)**：頻率越高，電流越集中在銅箔表層，損耗急劇攀升。

在傳統可插拔光模組 (Pluggable Optical Transceiver) 架構下，交換晶片 (Switch ASIC) 位於主機板中央，高速電信號必須經由 PCB 走線一路延伸到前面板 (Front Panel) 的光模組插槽，走線長度通常達 10~18 英吋 (約 25~45 公分)。為了確保信號到達前面板時通道損耗不超過規範 (如 IEEE

802.3ck 規範的 ~28dB 至 32dB 總預算)，整個主機板必須全面採用超低介電損耗的 M8、M9 等級 CCL 銅箔基板。



Source: TSMC

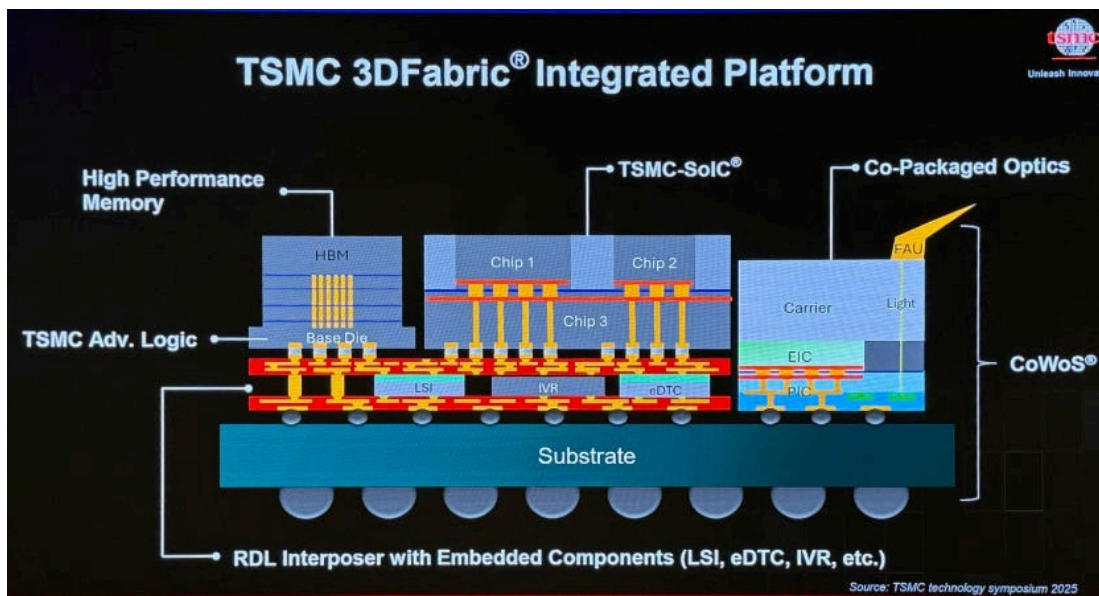
付費訂閱戶可分享光纖通訊轉 Co-Packaged Optics 及 Optics on Interposer 不需要用 M8/M9 銅箔基板的理由，對產業鏈的實質影響何時發生，誰受惠，誰受害及結論

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2. CPO 及 OOI 架構中移除了 PCB 上的長距離高速電信號通道

在 CPO 及 OOI (Optics on Interposer) 架構中，光引擎（Optical Engine, OE）直接封裝在與 Switch ASIC 相同的基板（Substrate / Interposer）上：

- **電-光轉換（E-O / O-E）提前完成：**電信號只在先進封裝載板（Substrate）內部走極短距離（通常僅幾毫米到 1~2 公分），隨即在光引擎內轉換為光信號。
- **離開基板即是光纖（Optical Fiber）：**從封裝邊緣拉到前面板或機櫃外部的全部是光纖纜線，而光纖在機房短距內的傳輸損耗幾乎可以忽略不計（光纖損耗每公里僅約 0.2~0.3dB，數十公分內接近 0dB）。



Source: TSMC

3. 主機板 PCB 的功能轉變為中低速與配電

當最嚴苛的 112G/224G 高速電路走線被限制在晶片載板內部後，底層主機板（Main PCB / Motherboard）所承載的信號發生了根本改變：

- **高速走線不再經過 PCB**：主機板上不再需要佈設長距離的 112G/224G SerDes 高頻差分走線。
- **PCB 僅需負責輔助功能**：主要傳輸低速控制訊號（PCIe Gen4/5、I2C、SPI、GPIO、帶外管理網絡）、時鐘信號（Clock）以及大規模電源供應（Power Delivery Network, PDN）。
- **M4 CCL 的性能已綽綽有餘**：對於中低頻與控制信號，M4 等級 CCL 的信號完整性（Signal Integrity）餘裕度已經非常充裕，完全不需要支付 M8/M9 高昂的材料與加工成本。

4. 成本與製造工藝的綜合考量

除了電氣性能，降階至 M4 亦大幅降低了生產門檻：

- **材料成本**：M8/M9 採用極低損耗樹脂體系（如 PTFE、改性 PPO/PPE）與超低輪廓銅箔（HVLP/HVLP2），成本常為傳統 M4（FR-4 / 中階改性環氧樹脂）的 3 到 5 倍以上。
- **PCB 製程良率**：M8/M9 壓合溫度高、膠片流動性控制嚴苛，多層厚板（40層以上）鑽孔磨損快且孔金屬化（PTH）難度極高；M4 的加工性、熱機械穩定性與量產良率顯著優於 M8/M9。

5. 核心結論：

傳統架構下主機板被迫採用 M8/M9，本質上是為了「用昂貴材料補償銅箔走線距離過長所產生的損耗」。CPO 直接從系統拓撲上消滅了 PCB 上的高

速電信號傳輸距離，原本壓在主機板上的損耗預算負擔直接解除，因此 PCB 材料退回 M4 即已完全符合電氣規範。

No near term impacts

從短期（2H26～1H28 年）來看，CPO 不會削弱 M8/M9 CCL 的需求，高階 CCL 的出貨量依然處於高速成長爆發期；但到了長期（2028 下半年以後，特別是 2029～2030 年），隨著 CPO 與光互連全面進入超大型 AI 叢集與交換機核心，M8/M9 在「主機板/交換機板」的單位面積需求確實會面臨結構性壓縮。

Why no near term impacts to M8/M9 demand ?

1. 傳統插拔模組在 800G / 1.6T 依然是絕對主流

- 當前 AI 伺服器機櫃（如 NVL72、各家 ASIC 叢集）與大型數據中心交換機（51.2T、102.4T）在 2026～2027 年仍以 Pluggable Optical Transceiver 為主。
- 在單通道 112G PAM4 及剛進入量產的 224G PAM4 架構下，只要走線仍留在 PCB 上，交換板（Switch Board）、運算加速卡（OAM/UBB）對 M8 / M9 等級 CCL 的滲透率反而正在急遽攀升，目前正處於規格升級帶來的量價齊揚階段。

2. 銅纜直連（DAC）與飛線（Flyover Cable）扮演過渡角色

- 在 CPO 成熟前，系統廠為了解決 PCB 走線過長損耗問題，大量採用「內部飛線/跳線（Internal PCIe/SerDes OverPass Cable）」或「銅纜背板（如 NVLink 銅纜架構）」。
- 這種做法雖然稍微減少了部分板層的極限走線，但系統主板（Baseboard）為了支撐密集的信號連接器與高頻 SerDes 扇出，依然維持在 M7/M8/M9 等級。

3. 高頻需求轉移至「載板（Substrate）」而非消失

- CPO 並沒有消除 112G/224G 的信號損耗問題，只是把電走線「從 PCB 搬到了先進封裝 ABF 載板或矽/玻璃中介層（ABF Substrate / Interposer）」。
- 原本 PCB CCL 廠商的高階產能，在供應鏈中會轉向超高層數載板用材料，或是被 AI 伺服器其他運算板（GPU Baseboard、OAM）吸收。

When the long term structural changes to happen?

時期	規格與架構特徵	CPO 滲透率	對 M8/M9 CCL 需求的實質影響
2025 ~ 2026	51.2T 交換機、800G/1.6T 插拔模組、NVL72 銅纜直連	< 1% (僅實驗性部署與小批量 PoC)	毫無負面影響。AI 伺服器全面升級 M8/M9，需求達到歷史新高。
2027 ~ 2028	102.4T 交換機、224G SerDes、Scale-up/Scale-out 光互連開始導入	約 5% ~ 15% (局限於頂級 CSP 的核心交換層)	邊際成長放緩。交換機板的 M8/M9 面積開始被 CPO 替代，但 AI 運算端 (GPU/ASIC 模組) 仍大量消耗高階 CCL。
2029 ~ 2030+	204.8T 交換機、448G SerDes、晶片間光互連 (Optical I/O) 成熟	> 30% ~ 50% (超大型叢集核心普及)	產生實質排擠效應。交換主板與部分 Scale-out 背板確定降階至 M4~M6，高階 CCL 成長動能將受壓制。

Source : Andrew Lu on global semis/techs

封裝型態 (Packaging Type)	放置位置 (OE Position)	傳輸介面 比重 (Cu vs. Optics)	頻寬密度 (Bandwidth Density)	能耗效率 (vs. 銅線)	延遲降低 (vs. 銅線)	量產預計時程 (Readiness)
Pluggable Optics (傳統可插拔)	PCB 邊緣插槽	100% 銅線 (PCB)	~800Gbps	1X	1X	已成熟量產
OBO (板上光學)	PCB 內部	80% 銅線 + 20% 光學	<1.6 Tbps	~2X (0.8x 能耗)	0.75X	已成熟量產
NPO (近封裝光學)	PCB 與 Substrate 邊界	50% 銅線 + 50% 光學	< 3.2 Tbps	~2.5X (0.6x 能耗)	0.50X	少量試產 (黃燈)
CPO on Substrate (COUPE 3.0)	ABF 載板 (Substrate)	20% 銅線 + 80% 光學	2 Tb/s/mm (<6.4T)	4X (0.25x 能耗)	<0.10X	2026 年小量產 (黃燈)
CPO on Interposer	CoWoS 中介層	10% 銅線 + 90% 光	10 Tb/s/mm (<12.8T)	10X (0.10x 能	<0.05X	預計 2030 年 (紅燈)

Source: TSMC

Positive and negative impacts to supply chain vendors

1. 潛在受衝擊端：高階高多層 PCB / CCL 廠商

- 目前一台高階 AI 交換機主板，往往採用 36~50 層的 M8/M9 PCB，單板價值極高。
- 若 CPO 在 2028 年後成為標配，交換機主板降階為 16~24 層的 M4 甚至普通 FR-4 板，主板的單機價值量 (ASP) 將暴跌 50%~70%。這將直接壓縮台燿 (TUC)、聯茂 (ITEQ)、台光電 (EMC) 以及美日材料龍頭 (Panasonic、Isola) 在高階網絡主板的成長天花板。

2. 價值轉移端：半導體先進封裝與光學零組件

原本 PCB 廠商賺取的「材料低損耗溢價」，將轉移給：

- **矽光子 (SiPh) 晶圓代工與封裝** (TSMC COUPE 等先進光學封裝技術)。
- **高階載板廠 (ABF Substrate)**：如欣興、Ibiden、南電，因為光引擎與 ASIC 整合封裝需要更大面積、更高精密度的載板。但要是未來光纖技術繼續轉向 OOI (Optics on Interposer)，ABF載板的重要性也相對會降低。
- **高精度光纖陣列連接器 (MT/MPO 等光纖跳線)**。

Conclusions

從投資或產業鏈的視角來看，未來 1-2年（至 2H28 年）無需擔憂 CPO 衝擊 M8/M9 需求，因為 224G 插拔模組與大型 AI 叢集的出貨量正在迎來陡峭爬坡期。真正需要警戒的時間窗口在 2028 年底至 2029 年（伴隨 102.4T / 204.8T 世代放量），屆時 CPO 若成功突破散熱、良率，標準化，及成本問題，高階通訊 PCB/CCL 的成長曲線才會真正面臨影響，但前提還是要關注 CPO及OOI 是否能在AI數據中心機櫃內 Scale up 中全面取代 GPU/CPU/ASIC 的銅線互聯。

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From: Yardeni QuickTakes <yardeni-research@ghost.io>
Subject: The Hawks Fly High Over a Resilient Economy
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YARDENI QUICKTAKES

Fed's Hawks Circling Over Hot US Economy

By Ed Yardeni & Elias Griepentrog • 16 Sept 2026

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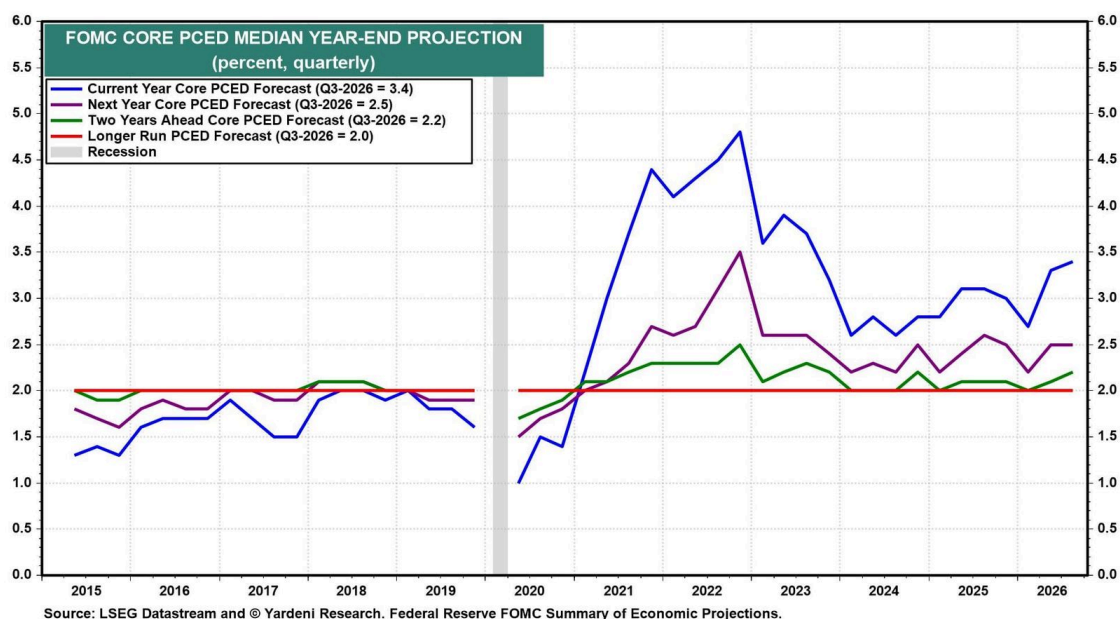
Photo by [Markus Kammermann](#) / [Unsplash](#)

I. The Fed

Today marked the conclusion of the September 15-16 FOMC meeting. The Fed's monetary policy committee delivered a widely anticipated 25bps increase in the federal funds rate (FFR), raising the target range to 3.75%-4.00%. Here are five key takeaways from today's decision:

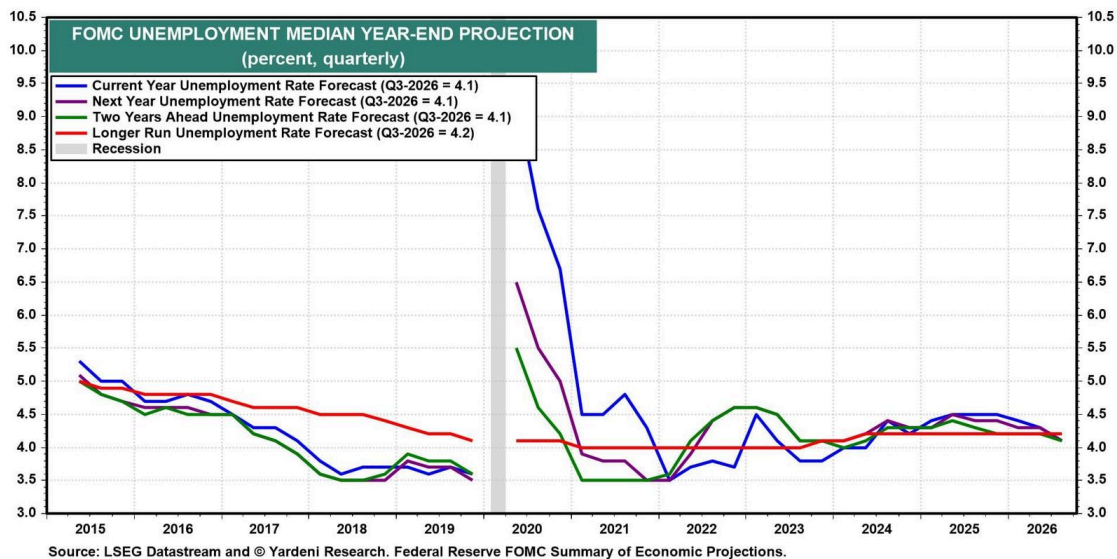
(1) The decision was unanimous. The FOMC voted 12-0 to raise the FFR by 25bps, showing unanimous agreement that tighter monetary policy is warranted. During his press conference today, Fed Chair Kevin Warsh said the vote “shows our resolve to achieve price stability on a timelier basis.” He pointed to three developments since July that brought the Committee together: stronger economic growth, insufficient improvement in inflation, and increased geopolitical risks. We reckon that the re-escalation of the war in the Middle East and the resulting prospect of more inflationary pressures from higher-for-longer oil prices was the deciding factor.

(2) Inflation remains the Fed’s predominant concern. Warsh said “inflation is too high and has been for too long” and that the Fed’s “predominant focus is on the price stability side of our mandate.” He added that this summer’s inflation readings “do not tell me that underlying trends have meaningfully improved,” with several key measures still running above 3.0% y/y. The Summary of Economic Projections (SEP) reinforced that message, with 2026 headline and core PCED forecasts revised slightly higher and inflation not forecast to return fully to the Fed’s target of 2.0% y/y until 2029 (chart).



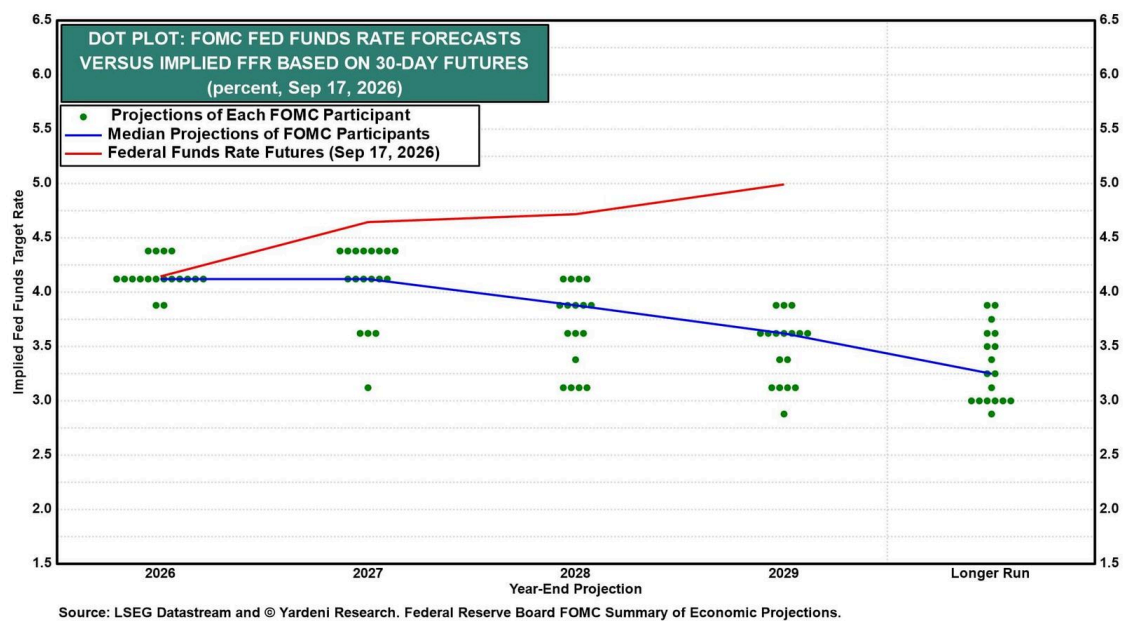
(3) The economy is stronger than the Fed thought in June. Warsh repeatedly emphasized that the “American economy appears to be strengthening,” pointing to improving hiring, private-sector earnings, business capital investment, and robust credit flows. He also said he would be “hard-pressed to describe broad financial conditions as restrictive,” a view widely shared across the Committee. The SEP similarly

revised growth modestly higher and unemployment lower to 4.1% through 2028 (chart). Warsh also characterized the labor market as essentially at full employment, saying the “labor side of the Fed’s congressional remit is in good shape.” That gives the Fed more room to focus on the inflation side of its dual mandate.



(4) The bar for another rate hike is low. The median of the 19 participants now expects another 25bps hike this year, no cuts in 2027, and only gradual easing thereafter, suggesting that today’s move was not intended as a one-and-done increase (chart). Four participants expect a third rate hike this year, while eight expect it in 2027.

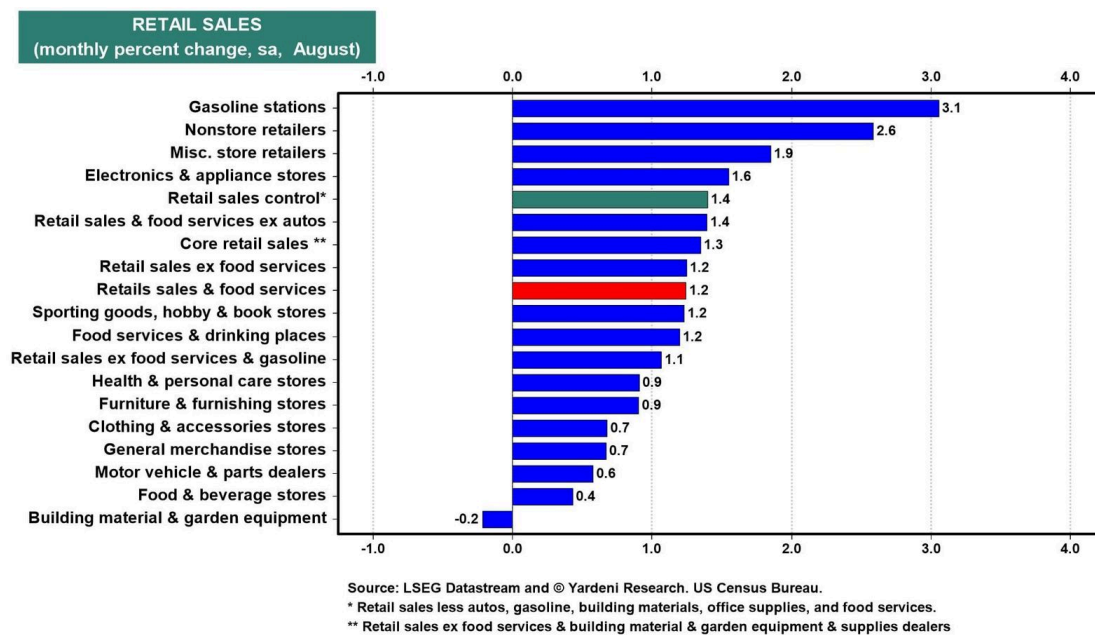
Warsh refused to pre-commit, saying “I’m not in the forward guidance business,” but his reaction function was clear. Underlying inflation must move toward 2.0% “clearly and at sufficient speed,” and he said today that “this standard has not been satisfied.” He also stressed that the Committee had merely “removed a dose of accommodation” and remains “hard-pressed” to describe financial conditions as “restrictive.” Unless inflation moderates clearly, the case for another hike remains intact while economic growth is strong, the labor market is near full employment, and financial conditions are not restrictive.



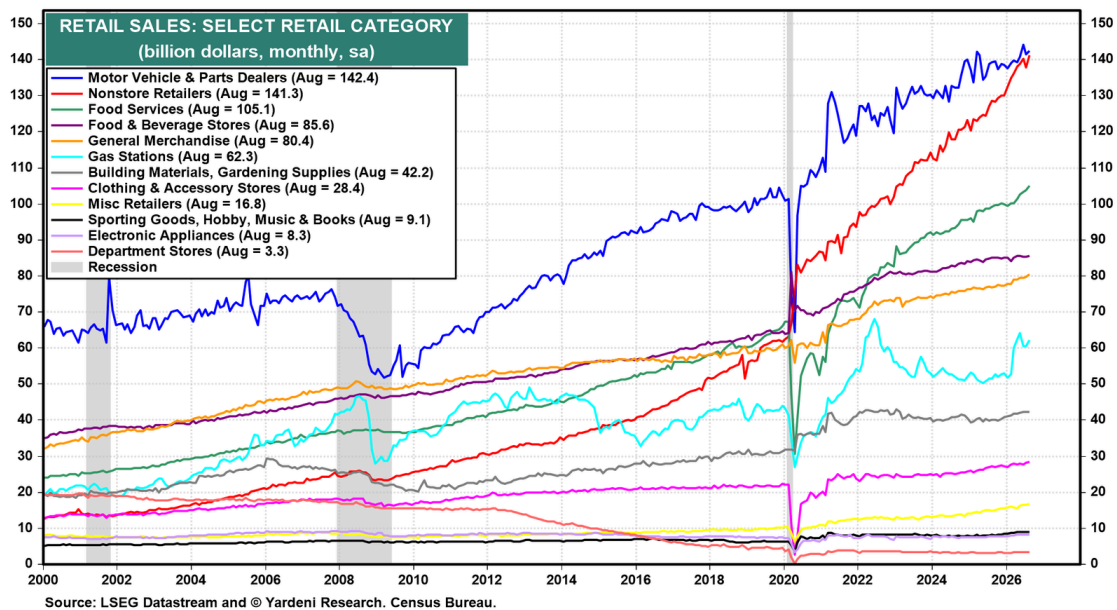
II. US Economy

As the Fed delivered a hawkish rate hike, the latest economic data reinforced both the remarkable strength of the economy and the persistence of inflation. Here's a look:

(1) Retail sales. August retail sales rose 1.2% m/m, above the 0.8% expected and the strongest gain since March 2026 (chart). Control-group sales, used in calculating GDP, surged 1.4% versus 0.5% expected. That was the strongest increase in nearly two years. The strength was broad-based, with 12 of 13 categories rising. Gasoline-station receipts jumped 3.1% as prices at the pump averaged about \$4.06 per gallon in August, yet discretionary spending remained strong: food services & drinking places rose 1.2%, the most since May, while sporting goods also increased 1.2%.

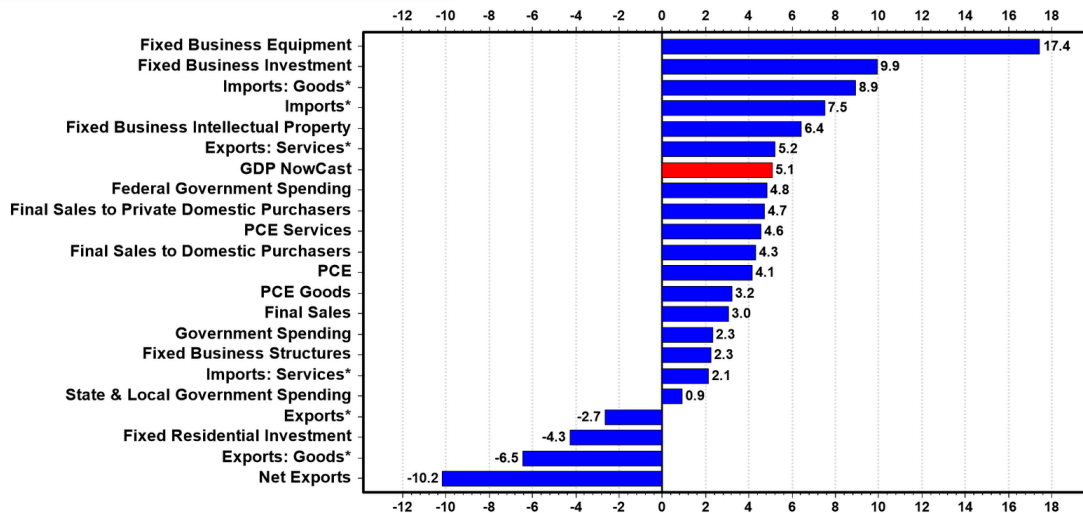


Nonstore retail sales jumped 2.6% m/m to a record high (chart).



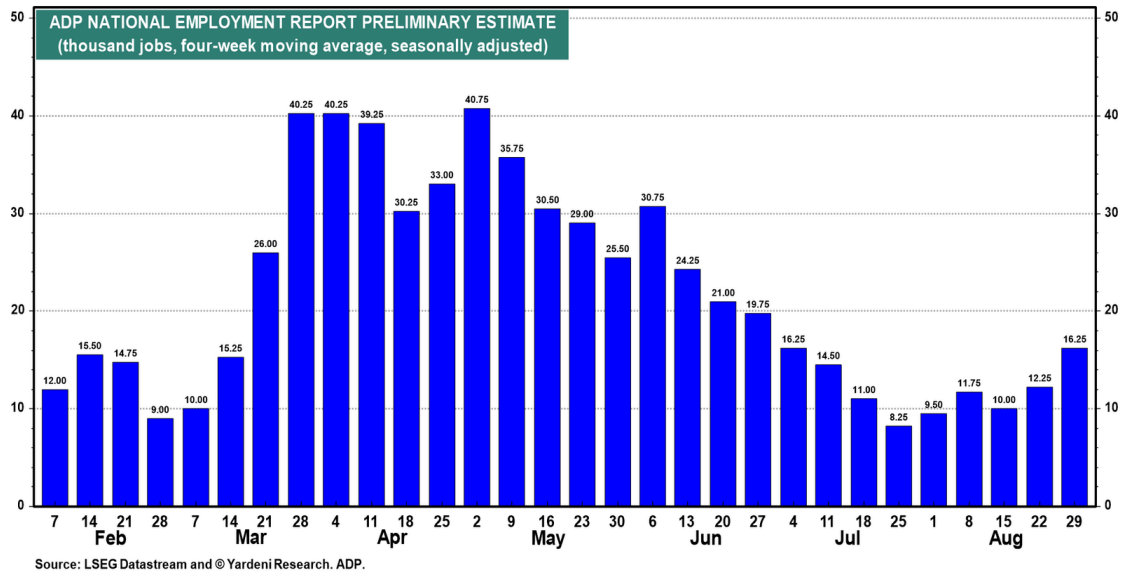
(2) GDP. The Atlanta Fed's [GDPNow](#) tracking model revised its Q3 real GDP growth estimate up from 4.4% to 5.1% (saar). The upgrade was driven largely by stronger consumer spending, with real PCE growth now tracking at 4.1%, up from 3.6% (chart). That would mark the strongest quarterly increase in consumer spending since Q1-2023. Business spending also remains robust, reinforcing the picture of an economy supported by both resilient consumers and strong capital spending.

ATLANTA FED GDPNOW ESTIMATE FOR Q2 2026
(quarterly percent change, saar, September 16)

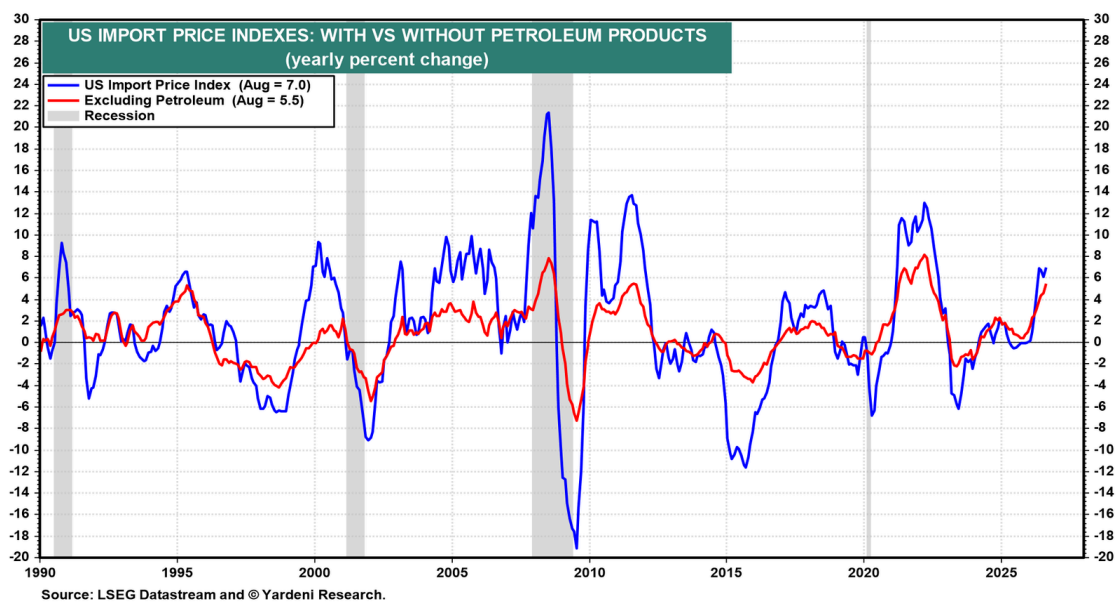


Source: LSEG Datastream and © Yardeni Research.
* Exports add to GDP, while imports subtract from GDP.

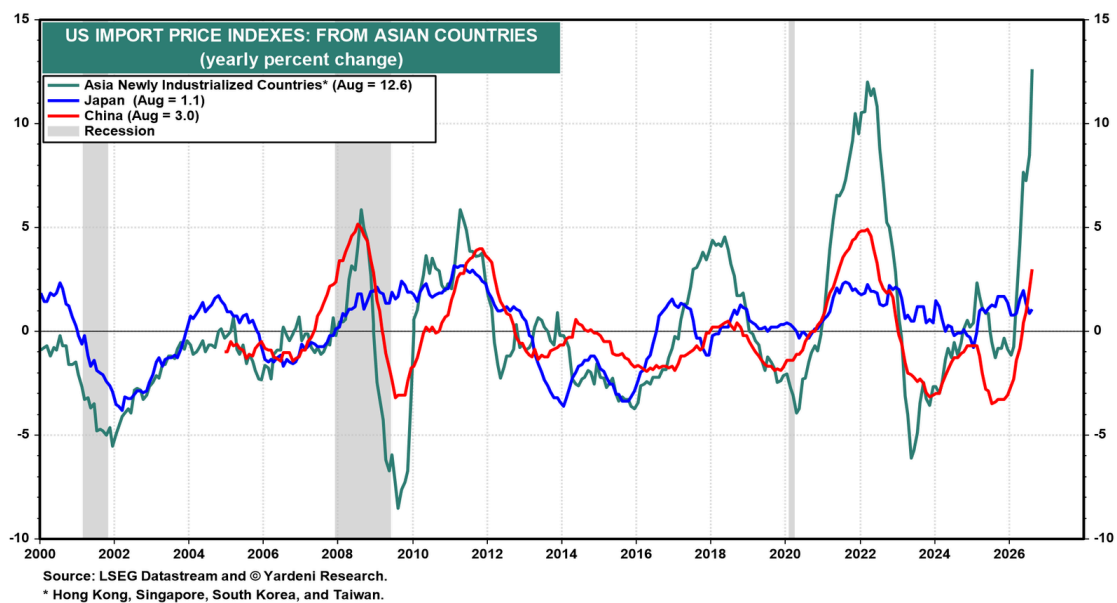
(3) ADP. A key reason for the resilience in consumer spending is the strength of the labor market. US private employers added an average of 16,250 jobs per week in the four weeks ending August 29, the most since early July (chart). That is consistent with a monthly pace of roughly 65,000 jobs, suggesting that the economy continues to operate at full employment.



(4) Import prices. August's data on import prices point to persistent inflation pressures. The import price index rose 7.0% y/y in last month, the fastest pace since August 2022. Petroleum import prices rose 27.3%. But even excluding petroleum, import prices rose 5.5%, the highest increase since May 2022 (chart).



Import prices from the newly industrialized Asian countries surged 12.6% y/y in August, reflecting AI-related demand for semiconductors, servers, memory, and other electronics outstripping supply (chart). That suggests the AI buildout will remain inflationary for now, before AI brings the fruits of disinflationary productivity growth.



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Subject: Alphabet and Blackstone's new JV gets US\$22bn bank loan, implications?
Date: 1 day ago

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Alphabet and Blackstone's new JV gets US\$22bn bank loan, implications?

谷歌追隨 Nvidia 的 AI GPU 資產化腳步，首次將其 TPU 及合約資產化，那誰沒跟上？

ANDREW LU ON GLOBAL SEMI/TECHS
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What's new?

據路透社報導（[Banks provide \\$22 billion chip loan to Blackstone, Alphabet AI cloud venture, source says](#)）十家全球銀行如高盛，Sumitomo Mitsui Banking Corp, 巴克萊銀行，BNP Paribas, Bank of Nova Scotia 據報向 Blackstone 與 Alphabet 合資成立的 JV Crux AI 提供 US\$22bn 晶片貸款（請參考我們之前的深度報告：[A new TPU Neocloud is emerging](#)），用於購買 Google TPU；貸款由 TPU 資產與 Crux AI 客戶合約擔保。Blackstone 先前宣布投入 US\$5bn 初始股本，目標 2027 年讓 500MW 資料中心容量上線。主要當事人均未即時回應，因此利率、期限、客戶、take-or-pay、TPU 世代與實際 IT load 尚未確認。

Positively affect what supply chain vendors and why?

Google TPU／Alphabet 最直接受惠，因自研 ASIC 從內部成本中心擴張為第三方雲端可融資資產。當然谷歌的 TPU 的產業鏈也會間接受惠，如設計統包服務廠商如博通，聯發科，及 Marvell、晶圓代工台積電、HBM、先進封裝、光通訊、交換器、伺服器 ODM、電力與液冷均獲得 500MW 的潛在需求。Blackstone 亦可用專案融資放大股本回報，而非完全以自有資金承擔建設。如果此案成功，我們相信更多的私募資金除了跟 Nvidia 合作之外，也會找谷歌合作，目前來看 Nvidia AI GPU 及谷歌的 TPU 將為主要兩家能夠將其 AI 晶片資產化的公司。



Source: Andrew Lu on global semis/techs

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Negatively affect what supply chain vendors and why?

純 NVIDIA GPU neocloud 及 AMD 的 GPU 客戶將面臨 Google 以「ASIC+雲+金融」整套輸出的競爭；沒有長期客戶合約、只能以晶片殘值融資的開發商，資金成本可能上升。若 TPU 專案取得較低租金，GPU 現貨出租與高毛利 inference pricing 可能受壓。但銀行則承擔 TPU 生態較封閉、客戶集中與技術迭代造成的殘值風險。

Our outside-the-box, non-consensus thinking

市場可能把 US\$22bn 解讀為 TPU 本身已有強二級市場；我們認為更可能相反：因 TPU 的可轉售性低於通用 GPU，銀行必須同時取得客戶合約擔保。換言之，這不完全類似 Nvidia 的晶片融資故事，而是加上合約現金流證券化。未來真正有定價權的不是擁有最多規劃 GW，而是能提供可執行 take-or-pay、晶片替換保證及軟體遷移安排的供應商。

Our advice to investors

我們不能將 US\$22bn 或客戶合約擔保額直接列為 AI Crux 收入。追蹤貸款期限／利率、客戶集中度、TPU 世代、500MW 的 IT-load/PUE、第一批上電節點及 Alphabet 是否提供回購／升級保證。供應鏈上偏好同時服務 GPU 與 ASIC 的 HBM、封裝、光通訊、ODM；對只依賴單一 TPU 世代的資產給較高殘值折價。

Risks to our analysis

這篇報導是二手消息且主要當事人未確認；US\$22bn 雖然代表目前在手訂單相當紮實，但可能含循環額度、設備或其他用途，未必全數動用。500MW 可能是園區 power 而非 IT load；客戶合約若極強，TPU 殘值風險的重要性可能低於本報判斷。

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Likely case Apple is stepping into enterprise AI server market?

蘋果用自研晶片配合 Nvidia NVLink Fusion 技術打算加入企業用AI伺服器市場

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What's new?

Reuters 引述 The Information 新聞報導 ([Apple Considers Return to Server Market, Has Talked With Nvidia to Use Network Tech](#))，Apple 正研究重新進入企業用伺服器市場，以 2 或 4 顆 M8 Ultra 晶片建構 AI 推理伺服器，並考慮採用 NVIDIA NVLink Fusion 連接 M8 chips。產品最早約 2029，目前仍可能取消或最終不採 NVIDIA；Reuters 無法獨立驗證。Apple 自2002年發表Xserve及2011 年停止 Xserve 產品後，若此案落地將是 18 年後重新出售 dedicated server hardware。

Exclusive

Apple Considers Return to Server Market, Has Talked With Nvidia to Use Network Tech

Any collaboration between the two tech giants would extend a nascent thaw after decades of frosty relations



Apple CEO John Ternus. Photo by Adam Gzag/Getty Images.

Source: The Information

Positive Impacts to supply chain vendors

NVIDIA：最重要。NVLink Fusion 的戰略價值是即使 AI 運算單元 compute die 不是 NVIDIA GPU，NVIDIA 仍可供應 scale-up connectivity 的技術。Apple 是極端垂直整合的公司，若連 Apple 都認為 NVIDIA NVLink Fusion 是最佳晶片網路連結選項，NVLink 的平台護城河很高。

付費訂閱戶可分享我們對於受惠及負面廠商分析，給投資人的建議及 Outside the box 非主流想法

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Apple：M-series silicon 可從消費性及專業工作站電腦延伸到企業用AI推理伺服器，增加蘋果晶片的可接觸市場 TAM，新產品營收增量動能，與攤銷龐大的設計成本及拉高獲利率。

TSMC：Apple 伺服器晶片及Nvidia NVLink Fusion 將增加台積電先進製程的需求，與大型CoWoS封測，高速 I/O 晶片需求。

Optics / networking ecosystem：2→4 顆晶片組合只是起點；若未來擴到 rack-scale, switch、retimer、AEC/optics 與 high-speed connectors content 會上升。

Who can be hurt the most?

傳統 x86 inference server CPU vendors：若 Apple 的 performance/W 速度及耗能優勢能移植到伺服器，部分 private/enterprise inference workload 可能被自研CPU晶片吸收，這對x86 伺服器CPU供應商 Intel/AMD 不利。

純 GPU-share bearish thesis 也會受挑戰：自研 ASIC/CPU 份額增加並不必然使 NVIDIA 營收下降，因 NVIDIA 可以從 NVLink Fusion 技術收費，但AMD AI GPU 的TAM就會受到影響。

Apple 自己的風險則是 enterprise software/support ecosystem：硬體性能不是伺服器市場唯一門檻；MLX、framework support、fleet management、service/SLA 都需建立。

Advice to investors

短期：這是 2029 年才會量產的產品，不應計入 2027-28 EPS。真正可交易的是 NVIDIA NVLink Fusion 的生態系擴充的估值是否有提升的空間。

2029 年後：應分別觀察 NVIDIA 的 AI GPU + Scale-up fabric + Scale-out networking + CUDA Software 四個核心競爭力，而非只估 AI GPU 的出貨量。

Outside-the-box / non-consensus thinking

已知事實：雖尚未定案，Apple 正考慮自研 M8 Ultra AI server + NVIDIA NVLink Fusion，對於自研ASIC 晶片屬於第一梯隊的蘋果而言，首次從企業用AI推理伺服器切入市場，我們覺得而且非常合理；

我們的推論：NVIDIA 最理想的終局不一定是阻止 ASIC/TPU 市場做大，而是讓所有自研 ASIC/TPU 都必須經過 NVIDIA fabric。

也就是從：Intel era：CPU owns platform → NVIDIA GPU era：accelerator owns platform → AI heterogeneous era：fabric owns platform。

如果 Apple、MediaTek ASIC、Meta MTIA、其他 hyperscaler XPU 都能透過 NVLink Fusion 接入 NVIDIA系統，NVIDIA 可能從「晶片市占」轉為 AI interconnect toll road。這會使 GPU 出貨量市場份額衰退與 NVIDIA 毛利市場份額變化脫鉤。

反向情境是 UALink / Ethernet scale-up 或 hyperscaler proprietary fabrics 成功標準化，NVLink 無法成為跨 vendor default；那麼 NVIDIA fabric premium 將被壓縮。

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YARDENI QUICKTAKES

WEEKLY WEBCAST: Worry List Is Growing More Worrisome

By Ed Yardeni • 16 Sept 2026

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We're still strong believers in our Roaring 2020s scenario, hinging on a productivity boom that strengthens economic growth even as it contains inflation. However, we've shaved the subjective odds that we ascribe to that scenario from 80% to 70% and now see a 30% chance that rising geopolitical and other risks could derail it. Today, Ed and Elias update our deepened worry list, detailing the concerning prospects, as well as reiterate the reasons that we remain bullish. We still believe that strong earnings growth will lift the S&P 500 to our 8400 price target by year-end.



Exclusive Early Access for Paid Members: Below, you'll find Dr. Ed's latest webcast. Paid members can enjoy immediate access to the video. This content will become available to the public at a later date—don't miss out on early insights, consider upgrading today!



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